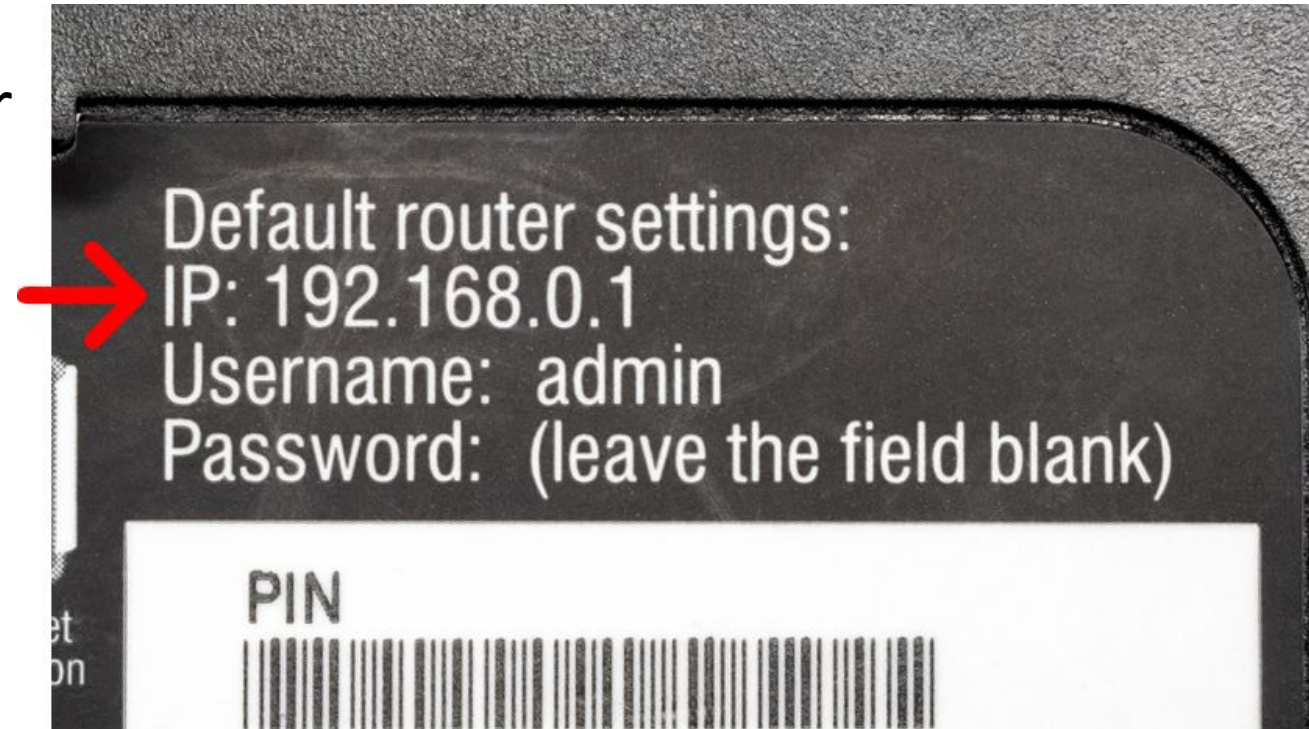


# Mobile Device Management & Networking (Brief 3)

Danyil Tymchuk NA  
for Mobile Tech module

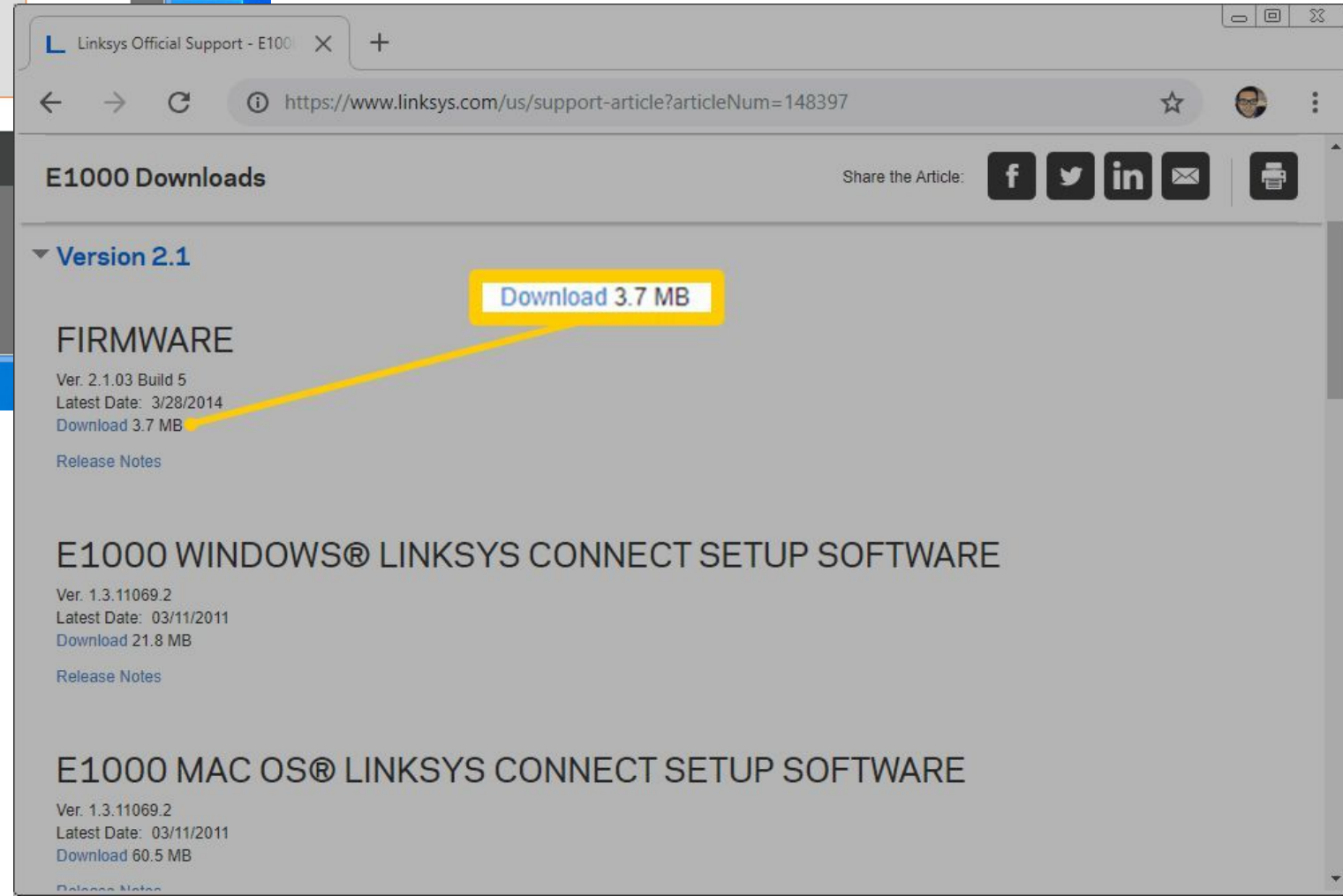
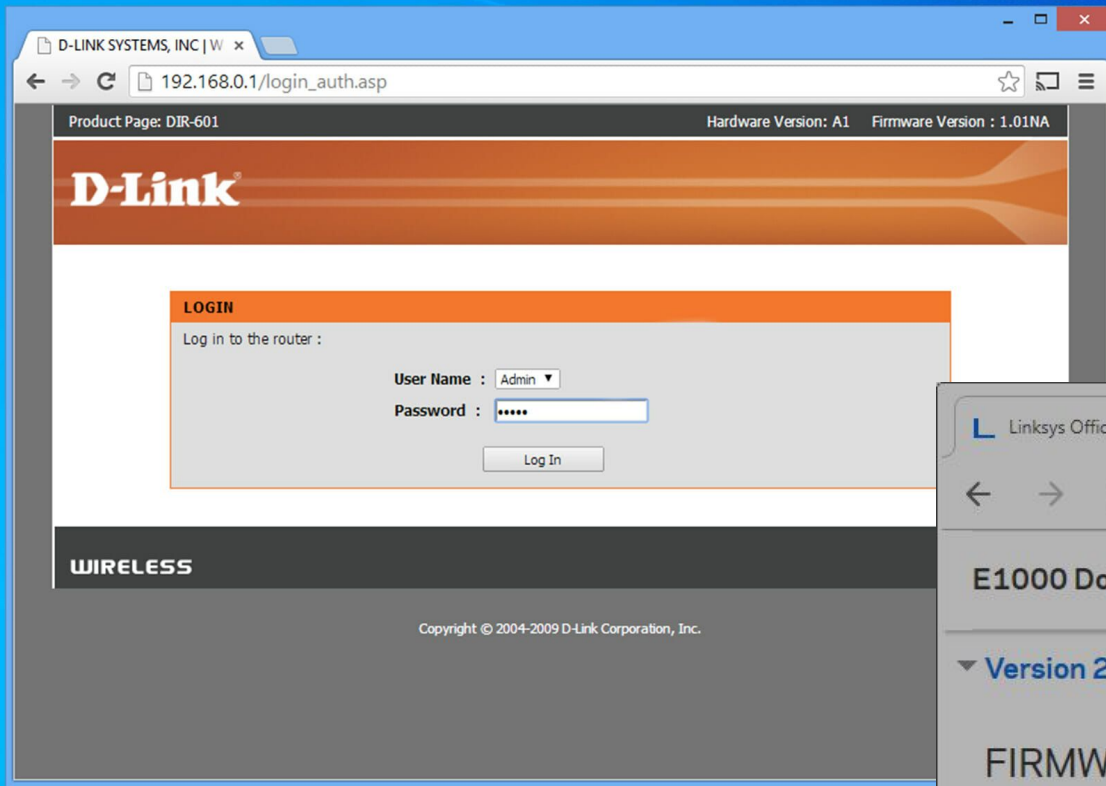
# Finding the Router Address and Default Gateway:

- The router address and default gateway were found on a sticker located underneath the router. The address was <https://192.168.0.1>

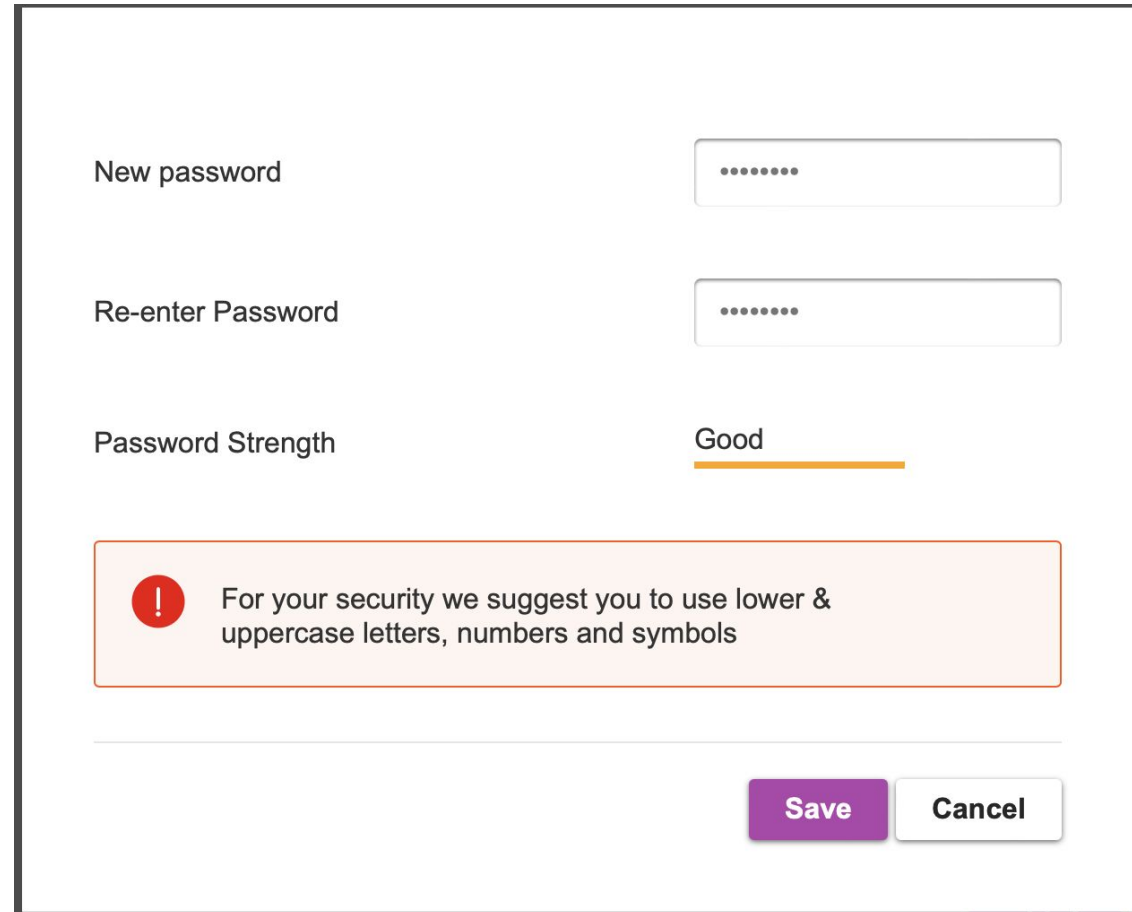


# Logging into the Router Admin Page and Checking for Firmware Updates

- Using the default credentials found on the sticker, the router admin page was accessed via a web browser. The firmware update section was navigated to, and the router was checked for any available updates.



## B. Change the default admin password



New password

Re-enter Password

Password Strength Good

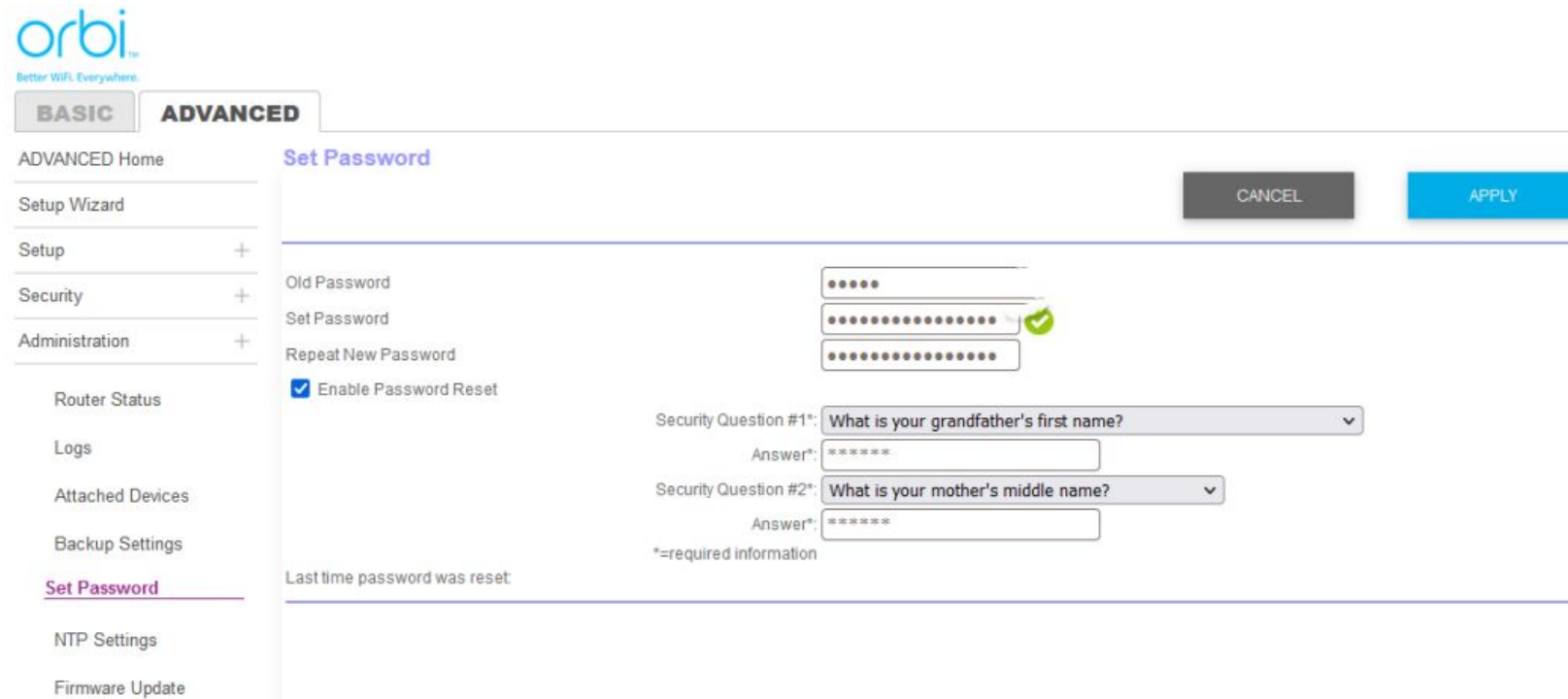
 For your security we suggest you to use lower & uppercase letters, numbers and symbols

**Save** **Cancel**

# Implementing Security Features:

- The following security features were implemented for the wireless router:
- a. The default admin password was changed to a strong, unique password. A screenshot of the password change process is included below.
- b. The WiFi network security was verified to be using WPA2. A screenshot of the security settings is included below.
- c. A MAC address filter was set up, and the device's MAC address was added to the allowed list. A screenshot of the MAC address filter settings is included below.
- d. SSID broadcast was disabled to hide the network. A screenshot of the setting is included below.
- e. The firewall was enabled, and all inbound traffic was blocked. A screenshot of the firewall settings is included below.
- f. ICMP pings were disabled on the public interface. A screenshot of the setting is included below.

# Set new password



The image shows the Orbi router's web interface. At the top left is the Orbi logo with the tagline "Better WiFi. Everywhere." Below it are two tabs: "BASIC" and "ADVANCED", with "ADVANCED" being the active tab. On the left side, there is a navigation menu with the following items: "ADVANCED Home", "Setup Wizard", "Setup" (with a plus icon), "Security" (with a plus icon), "Administration" (with a plus icon), "Router Status", "Logs", "Attached Devices", "Backup Settings", "Set Password" (highlighted in purple), "NTP Settings", and "Firmware Update". The main content area is titled "Set Password" in purple. It contains three password input fields: "Old Password" (filled with dots), "Set Password" (filled with dots and a green checkmark), and "Repeat New Password" (filled with dots). Below these is a checkbox labeled "Enable Password Reset" which is checked. To the right of the password fields are two "Security Question" dropdown menus. The first is labeled "Security Question #1\*" and has the text "What is your grandfather's first name?" selected. Below it is an "Answer\*" field filled with dots. The second is labeled "Security Question #2\*" and has the text "What is your mother's middle name?" selected. Below it is another "Answer\*" field filled with dots. At the bottom left of the main area, it says "Last time password was reset:". At the top right of the main area, there are two buttons: "CANCEL" (grey) and "APPLY" (blue). A small asterisk note at the bottom right says "\*=required information".

orbi  
Better WiFi. Everywhere.

**BASIC** **ADVANCED**

ADVANCED Home

Setup Wizard

Setup +

Security +

Administration +

Router Status

Logs

Attached Devices

Backup Settings

**Set Password**

NTP Settings

Firmware Update

### Set Password

CANCEL APPLY

Old Password

Set Password

Repeat New Password

☒ Enable Password Reset

Security Question #1\*: What is your grandfather's first name?

Answer\*:

Security Question #2\*: What is your mother's middle name?

Answer\*:

\*=required information

Last time password was reset:

# Set WPA2

The image shows a router's configuration interface. On the left is a sidebar menu with the following items: 'Wireless 2.4GHz' (highlighted in green), '- Wireless Settings', '- WPS', '- Wireless Security' (highlighted in green), '- Wireless MAC Filtering', '- Wireless Advanced', '- Wireless Statistics', and 'Wireless 5GHz' (highlighted in green). The main content area is titled 'Wireless Security' and contains two radio buttons: 'Disable Security' and 'WPA/WPA2 - Personal(Recommended)'. The 'WPA/WPA2 - Personal(Recommended)' option is selected. Below this, there are three fields: 'Version' set to 'Automatic(Recommended)', 'Encryption' set to 'AES', and 'PSK Password' set to '12345670'. A red box highlights the 'WPA/WPA2 - Personal(Recommended)' section and its fields. Below these fields is a note: '(You can enter ASCII characters between 8 and 63)'. At the bottom, there is a 'Group Key Update Period' field set to '0' with the text 'Seconds (Keep it default if you want)'.

**Wireless 2.4GHz**

- Wireless Settings
- WPS
- **Wireless Security**
- Wireless MAC Filtering
- Wireless Advanced
- Wireless Statistics
- Wireless 5GHz**

☐ Disable Security

☒ **WPA/WPA2 - Personal(Recommended)**

Version: Automatic(Recommended) ▼

Encryption: AES ▼

PSK Password: 12345670

(You can enter ASCII characters between 8 and 63)

Group Key Update Period: 0 Seconds (Keep it default if you want)



# Set MAC address filter

**ARP List**

| ID | MAC Address       | IP Address   | Status | Configure                                   |
|----|-------------------|--------------|--------|---------------------------------------------|
| 1  | 00-19-66-CA-8B-C7 | 192.168.1.18 | Bound  | <a href="#">Load</a> <a href="#">Delete</a> |

Bind All

Load All

Refresh

**Wireless MAC Filtering**

Wireless MAC Filtering: Disabled 

Enable

**Filtering Rules**

☐ Deny the stations specified by any enabled entries in the list to access.

☒ Allow the stations specified by any enabled entries in the list to access.

| ID | MAC Address       | Status  | Description | Modify                                        |
|----|-------------------|---------|-------------|-----------------------------------------------|
| 1  | E0-05-C5-84-20-89 | Enabled | home        | <a href="#">Modify</a> <a href="#">Delete</a> |

Add New...

Enable All

Disable All

Delete All

**Add or Modify Wireless MAC Address Filtering entry**

MAC Address: 

00-19-66-CA-8B-C7

Description: 

Wireless MAC Filter One

Status: 

Enabled

Save

Back

**Wireless MAC Filtering**

Wireless MAC Filtering: Disabled 

Enable

**Filtering Rules**

☐ Deny the stations specified by any enabled entries in the list to access.

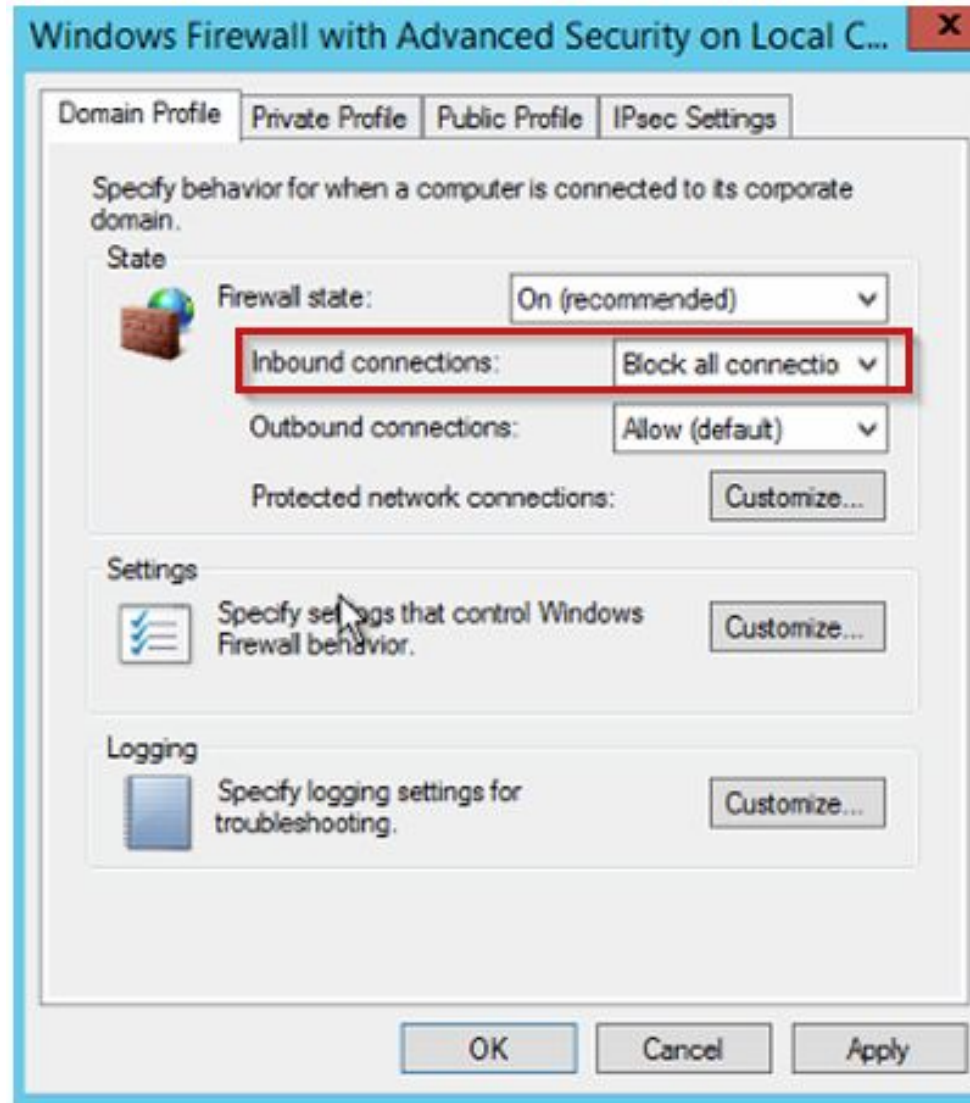
☒ Allow the stations specified by any enabled entries in the list to access.

# SSID broadcast

The image shows a router's web interface for configuring wireless settings. The left sidebar contains a menu with the following items: Quick Setup, Network, Dual Band Selection, Wireless 2.4GHz, - Wireless Settings (highlighted with a red circle and a red arrow pointing to it), - WPS, - Wireless Security, - Wireless MAC Filtering, - Wireless Advanced, - Wireless Statistics, Wireless 5GHz, Guest Network, DHCP, and USB Settings. The main content area is titled 'Wireless Settings (2.4GHz)' and contains the following fields and options:

- Wireless Network Name:  (Also called the SSID)
- Mode:
- Channel Width:
- Channel:
- ☐ Enable SSID Broadcast (highlighted with a red box and a red arrow pointing to it)
- ☐ Enable WDS Bridging
- (highlighted with a red box and a red arrow pointing to it)

# Firewall



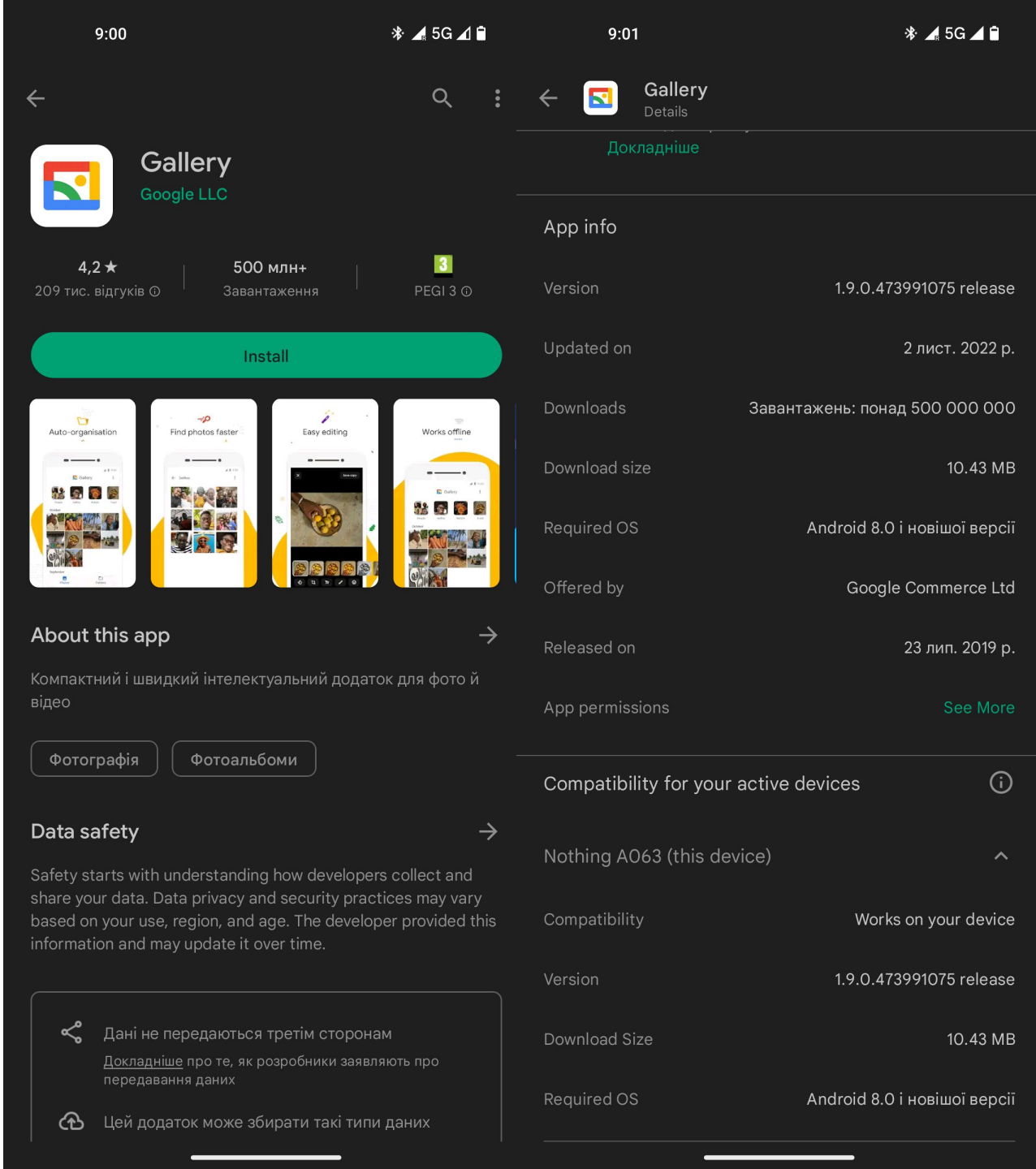
# ICMP pings were disabled on the public interface

---

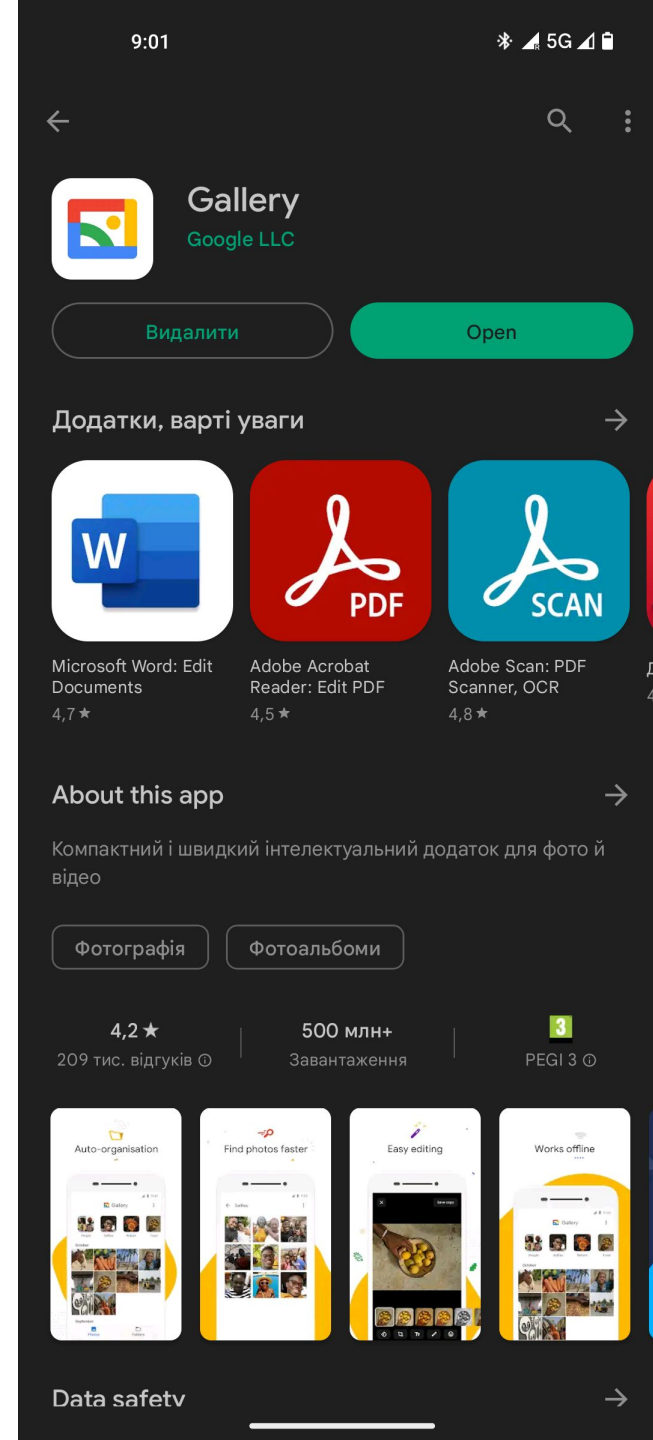
**Allow Ping to WAN interface**



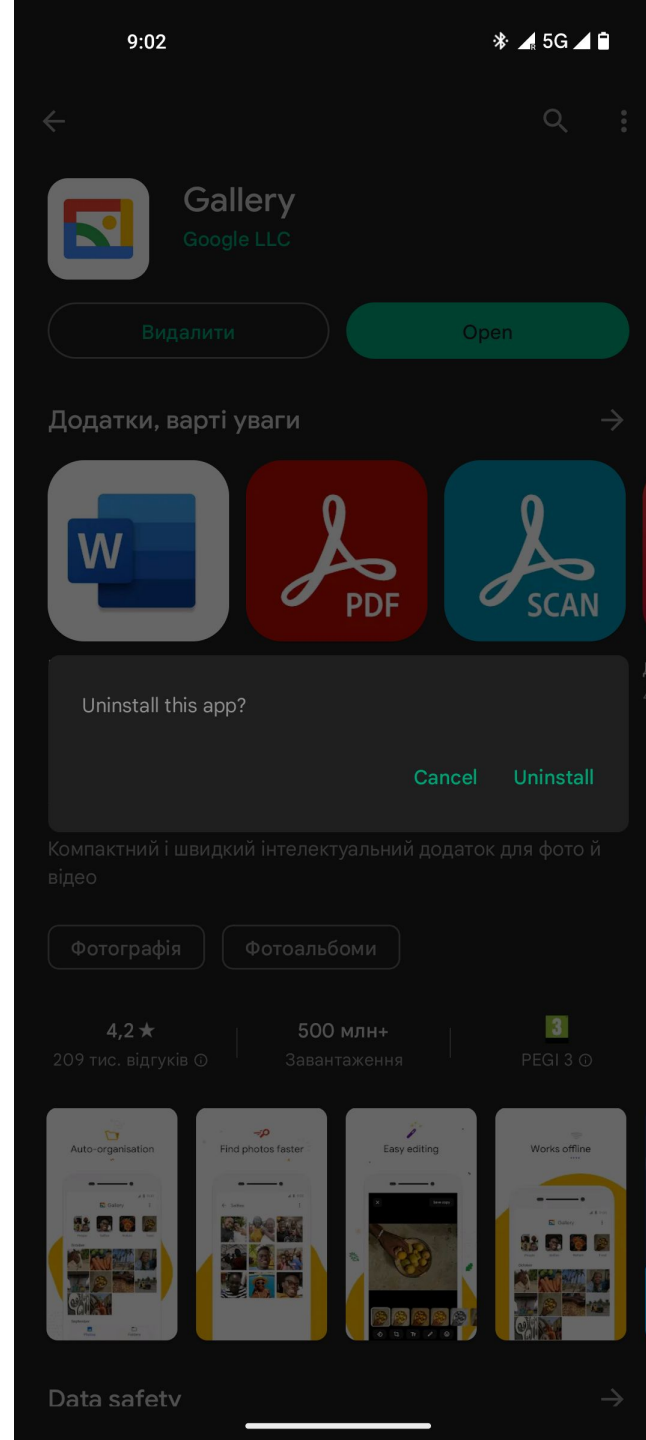
An app was chosen for installation from the Google Play Store. The app page, including hardware requirements, memory footprint, and supported OS versions, was captured in a screenshot.



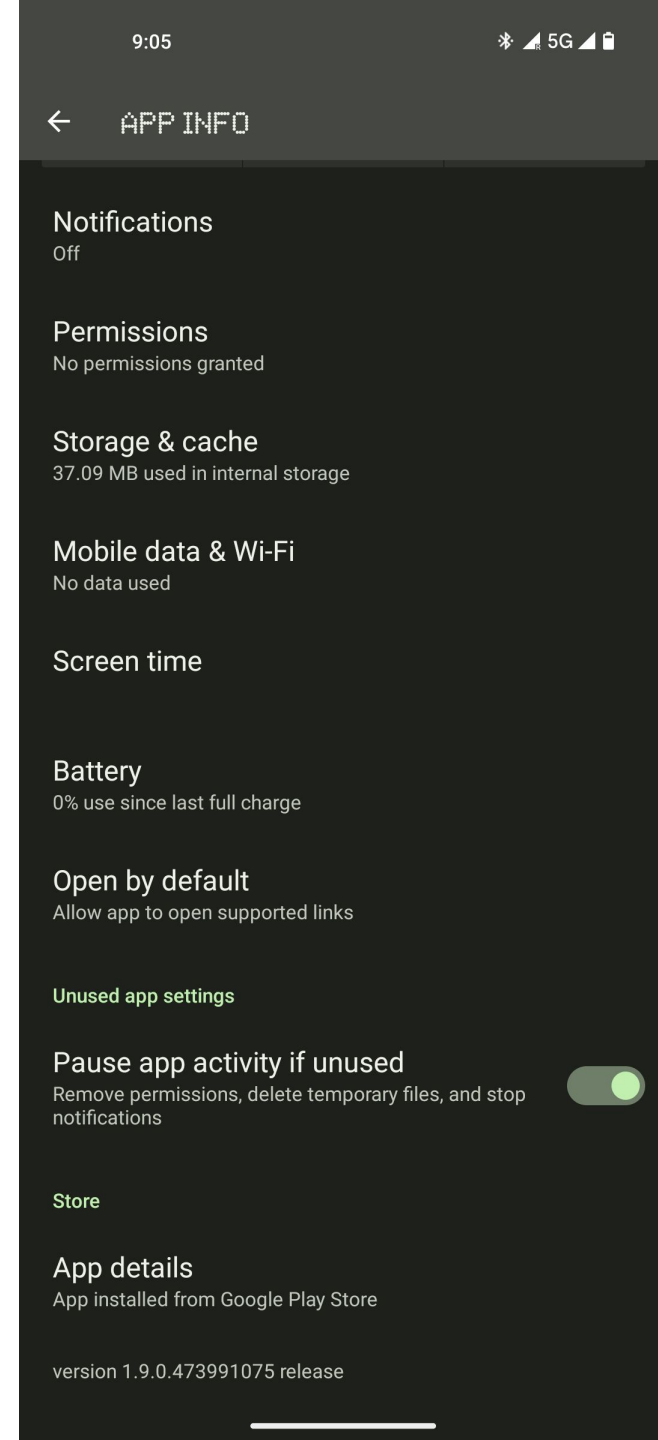
The app was then installed, and screenshots of the installation process were taken.



The app was subsequently uninstalled, and screenshots of the uninstallation process were taken.



The mobile device's settings were accessed to locate the software update section. A screenshot of the current software version and available updates was taken.





# Conclusion

- In conclusion, this report demonstrates the successful configuration of a wireless router with appropriate security features and mobile device management, as well as app installation and system update installation on a mobile device. The accompanying screenshots provide visual confirmation of the processes completed.

## Note:

- I don't have a wifi router at home, so I used photos from the internet